IC WebSocket Integration Guide

Insights copilot now provide websocket based real-time streaming of intermediate steps during conversation processing. This allows client applications to display insights as they are generated, creating a more interactive user experience.

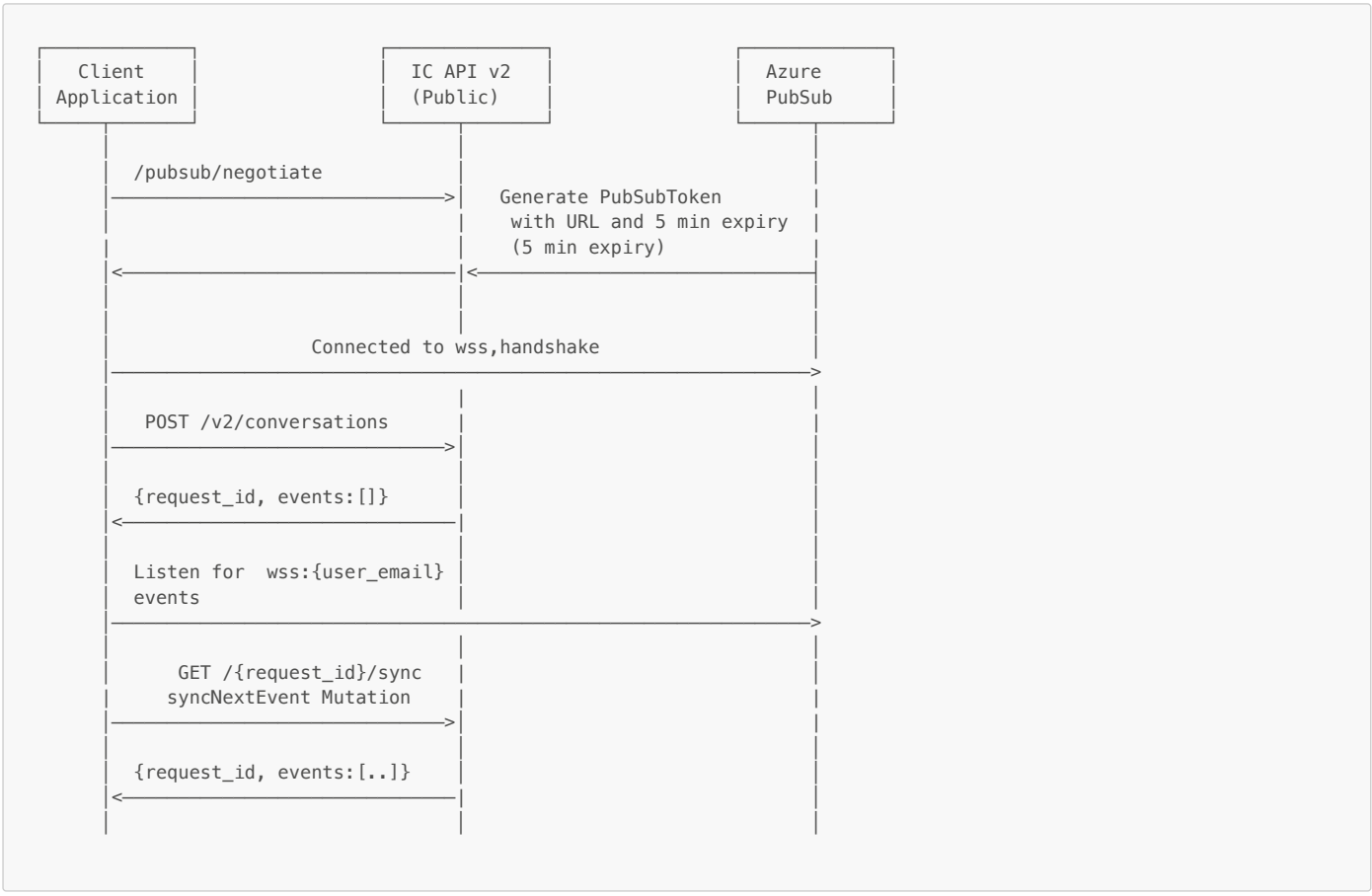
See live action of websocket with IC Web UI

Types of Events

The complete conversation processing is broken down into discrete steps, each represented as an event streamed over the WebSocket connection:

- planner: The initial plan generated by the planner tool.
- similar_plan_tool: Similar plans retrieved from the knowledge base.
- tool-executor: The execution of a specific tool, which can vary across analysis template, entity recognizer, python, chart_creation etc
- evaluator: The evaluation of tool outputs against the question.
- final-answer-generator: The generation of the final answer based on all previous steps.

Flow Diagram



Token-Based Authentication

IC issues **short-lived access tokens** embedded in the WebSocket URL. The Public API handles token generation automatically when you request streaming.

```
{
    "baseUrl": "wss://brewdatgbgaighqtechcopilotpubsub01d.webpubsub.azure.com/client/hubs/<hubname>",
    "token": "xxxx",
    "url": "wss://brewdatgbgaighqtechcopilotpubsub01d.webpubsub.azure.com/client/hubs/<hubname>?access_token=xxxx"
}
```

- Default token expiry is 5 minutes, which balances security and usability. Regardless of token expiry, the WebSocket connection remains active until explicitly closed or disrupted by network issues.

IC APIs to interact with WebSocket

IC API Docs

1. GET /pubsub/negotiate for WebSocket Connection Details

```
curl -X GET "https://api.insights-copilot.ab-inbev.com/v2/conversations/pubsub/negotiate" \
  -H "Authorization: Bearer <ACCESS_TOKEN>" \
  -H "Accept: application/json"
```

```
{
    "baseUrl": "wss://brewdatgbgaighqtechcopilotpubsub01d.webpubsub.azure.com/client/hubs/<hubname>",
    "token": "xxxx",
    "url": "wss://brewdatgbgaighqtechcopilotpubsub01d.webpubsub.azure.com/client/hubs/<hubname>?access_token=xxxx"
}
```

Connect to the provided `url` using a WebSocket client to start receiving real-time events. Azure Web PubSub SDK provides **automatic reconnection** for network interruptions:

```
import { WebPubSubClient } from "@azure/web-pubsub-client";
const client = new WebPubSubClient(url, {
  autoReconnect: true  // Automatically reconnect on network drops
});
client.on("message", (event) => {
  const data = JSON.parse(event.data);
  // Handle incoming event data
});
```

2. POST /v2/conversations or POST /v2/conversations/{conv_id} for QnA

Continue with the QnA apis as usual to trigger the conversation processing. The intermediate steps will be streamed back via the WebSocket connection.

```
curl -X POST "https://api.insights-copilot.ab-inbev.com/v2/conversations" \
  -H "Authorization: Bearer <ACCESS_TOKEN>" \
  -H "Content-Type: application/json" \
  -d '{
  "question": "What is the sales performance of Budweiser in the US last quarter"
}'
```

Custom Configurations adapted by IC UI for Optimal Display

IC UI uses the intermediate `task_response` for dynamically mapping the intermediate steps to user-friendly primary text and details sections.

Custom Mapping of Primary Text and Details

Use the `next_tool` from `task_response` to perform the mapping below.

Next_tool	Primary Text	Detail	Schema Key
(initial — no step)	"Thinking..."	NA	NA
planner	"Thinking about how to address your query..."	"Here's my initial plan: [X]"	X=outcome.plan
similar_plan_tool	"Planning next steps..."	"Found X similar questions I've answered before."	X = len(outcome)
similar_plan_tool	"Planning next steps..."	"This is a new type of question - building a fresh approach"	NA

Next_tool	Primary Text	Detail	Schema Key
planner _(refined)	"Analyzing your question... {Y}" OR "Creating your chart {Y}" OR "Running your Smart Flow" OR "Querying your data" OR "Crunching the numbers" OR "Updated plan:{X}" OR Looking up relevant context {Y}" OR "Searching internal reports {Y}", "Searching the web {Y}"	Updated plan: X	X = outcome.plan, Y = outcome.question
entity_recognizer_for_analysis_node	"Running your Smart Flow...".	"Identified from your question: {X} \nThis usually takes about 30 seconds"	X=outcome.extracted_entities/validated_entities/suggested_entities/data_ambiguity_sugge
analysis_tool *(not HIL)*	"Reviewing results..."	"Insights generated using Smart Flow: X"	X=outcome.stored_proc
analysis_tool _(HIL)	"Just a quick clarification..."		
evaluator (decision:conmplete)	"My review is complete!"	I have all the information I need to answer your question.	NA
evaluator (decision:incomplete)	"Improving your answer..."	Needs more work {X}	X =outcome.reasoning.split()[0]
chart_creation_tool	"Reviewing results..."	"Creating chart: {X} Chart complete ✓"	X = outcome[0:80]
doc_qa_tool	"Reviewing results..."	"Searching documents: {X} Internal document results:[Y] Sources: [filename]".	X=outcome[0:80]

Next_tool	Primary Text	Detail	Schema Key
internet_tool	"Reviewing results..."	"Searching the web: [First 80 chars of question...] Web search results:[tool output, truncated] Sources: [urls]"	X=outcome[0:80]
sql_tool	"Reviewing results..."	"Querying: [First 80 chars of question...] Results: [tool output, truncated] Table: [table_name]"	
business_knowledge_tool	Reviewing results..."	"Looking up: [First 80 chars of question...] Context retrieved:[tool output, truncated]"	X=outcome[0:80]
final_answer_tool	"Complete!"	Your response is ready	NA

NOTE: Custom mapping for entities:

- Symbols: no symbol = validated_entities
- strikethrough = len(suggested-entities)=1/unknown_entities,
- ? = suggeted_entitites/data_ambiguity

3. Output Length Management

As the tool outputs are LLM-generated, their length cannot be reliably controlled through prompting alone.

- Max display: First 3 bullet points or ~500 characters, whichever comes first
- Truncation indicator: Append "... *[details in full response]*" when truncated
- Full content: Always available in the Received Input JSON (for debugging) and in the final answer (for users)

Event Schema

Every intermediate step in the conversation processing follows below base schema. The `outcome` field contains tool-specific results, which can be customized for optimal display in the UI clients.

Base schema

```
{
    "node": "<Unique name of the tool node executed>",
    "question": "<[Optional]the question/task being processed>",
    "outcome": { /* tool-specific object */ },
    "langfuse_trace_id": "[Optional]<unique trace ID for this tool execution>",
    "next_tool": "<name of the next tool to execute>",
    "next_tool_timeout": 60
}
```

- **next_tool**: String indicating which tool will execute next in the workflow. Must match one of the standardized tool names (e.g., "planner", "analysis_tool", "chart_creation_tool").
- **next_tool_timeout**: Integer representing the expected maximum wait time in seconds before the next tool should respond. This allows the UI clients to detect dropped connections and show appropriate error states.
- **langfuse_trace_id**: Unique identifier for tracing the execution of this tool in Langfuse. This is critical for debugging and monitoring.

- **output** schema varies based on the tool type.

1.planner schema example


```
{
    "node": "planner",
    "question": "What are the drivers of Salience for Corona in Brazil?",
    "langfuse_trace_id": "c713b2f2243a835d245208b474ea2210",
    "next_tool": "similar_plan_tool",
    "next_tool_timeout": 15
    "outcome":
    {
        "plan": "1. Search for similar past plans regarding drivers of Salience for Corona in Brazil to leverage
proven approaches.\n2. If no relevant plan exists, retrieve detailed analysis of Salience drivers for Corona in
Brazil using analysis_tool.\n3. Review the analysis for key drivers and insights.\n4. Synthesize findings into a
comprehensive final answer"
        "reasoning": "The user is asking for the drivers of Salience for Corona in Brazil, which is a business
analysis question. Per guidelines, the first step is to check similar_plan_tool for any relevant past plans or
analyses to ensure consistency and efficiency. If no relevant plan is found, the next step will be to retrieve the
analysis using analysis_tool.",
    },
}
```


2.similar_plan_tool with plans


```
{
    "node": "similar_plan_tool",
    "question": "What are the Drivers of Salience for Corona in Brazil?",
    "next_tool": "planner",
    "next_tool_timeout": 15
    "outcome":
    [{'question': 'Trend of meaningfulness across years for Corona in Brazil',
    'overall_plan': '1. Retrieve yearly meaningfulness scores for  Corona in Brazil over the past years.\\n2.
Summarize and visualize the trend.','next_tool': 'analysis_tool', 'next_question': 'What are the yearly
meaningfulness for Corona in Brazil over the past several years?',
    'reasoning': 'The user wants a multi-year time series for one attribute and brand-country combo.'},
    {'question': 'Why did salience drop for Corona in 2024?', 'overall_plan': '1. Extract drivers of salience change
for Corona in 2024 and identify the main factors causing the drop.\\n2. Summarize findings and give
recommendations.', 'next_tool': 'analysis_tool', 'next_question': 'Why did salience drop for Corona in 2024?',
'reasoning': 'Causality question\u2014need to look for linked changes in performance or drivers for the drop.'}],

}
```


3. similar_plan_tool without plans


```
{
    "node": "similar_plan_tool",
    "question": "What are the Drivers of Salience for Corona in Brazil?",
    "next_tool": "planner",
    "next_tool_timeout": 15
    "outcome": [],

}
```


4. planner (refined)


```
{
  "node": "planner",
 "next_tool": "entity_recognizer_for_analysis_node",
  "langfuse_trace_id": "c713b2f2243a835d245208b474ea2210",
  "question": "What are the drivers of Salience for Corona in Brazil?",
  "next_tool_timeout": 60
  "outcome":
  {"plan": "1. Retrieve detailed analysis of the drivers of Salience for Corona in Brazil using analysis_tool.\n2.
Review the analysis for key drivers and insights.\n3. Synthesize findings into a comprehensive final answer.",
  "reasoning": "The user is asking for the specific drivers of Salience for Corona in Brazil. There is no indication
that this information has already been retrieved, and the similar_plan_tool did not return a directly matching plan.
Therefore, the next step is to use analysis_tool to obtain a detailed analysis of Salience drivers for Corona in
Brazil."
```

```
  },
  }
```


5. entity_recognizer_for_analysis_node


```
{"node": "entity_recognizer_for_analysis_node",
 "next_tool": "analysis_tool",
 "langfuse_trace_id": "c713b2f2243a835d245208b474ea2210",
 "question": "What are the Drivers of Salience for Corona in Brazil?",
  "outcome":
    {"human_question": "What are the drivers of Salience for Corona in Brazil?",
    "extracted_entities": { "KPI": "SALIENT", "BRAND": "CORONA", "COUNTRY": "BRAZIL"},
    "validated_entities": {"BRAZIL": ["COUNTRY"], "SALIENCE": ["KPI"], "CORONA EXTRA": ["BRAND"]},
    "suggested_entities": {"SALIENT": [{"suggested_value": "SALIENCE", "column": "KPI", "similarity_score": 86}],
"CORONA": [{"suggested_value": "CORONA EXTRA", "column": "BRAND", "similarity_score": 100}]},
    "data_ambiguity_suggestions": {},
    "unknown_entities": [], "hil_trigger": false, "ner_augmented_question": "what are the drivers of kpi salience for
CORONA EXTRA in country brazil?", "suggested_questions": ["what are the drivers of kpi salience for CORONA EXTRA in
country brazil?"], "request_id": "2460491179204099294", "response": null, "merged_unique_values": null,
"ambiguous_columns_mapping": null, "hierarchies": null, "tables_list": null, "unique_values_json": null,
"merged_unique_values_dict": null},
    "next_tool_timeout": 60
    }
```


6. analysis_tool


```
  {
  "node": "analysis_tool",
  "next_tool": "evaluator",
  "question": "what are the drivers of kpi salience for CORONA EXTRA in country brazil?",
  "outcome": {"stored_proc": "get_brand_highlights", "arguments": {"brand": ["CORONA EXTRA"], "country": ["brazil"],
"analysis_requested_for_kpi": ["salience"], "input_query": "What are the drivers of salience for CORONA EXTRA in the
country Brazil?""request_id": "2460491179204099294"}},
  "next_tool_timeout": 60
  }`
```


7. evaluator (complete)


```
{"node": "evaluator", "langfuse_trace_id": "c713b2f2243a835d245208b474ea2210", "question": "What are the Drivers of
Salience for Corona in Brazil?", "next_tool": "final_answer_tool", "outcome": {"decision": "complete", "reasoning":
"The user's question was: 'What are the Drivers of Salience for Corona in Brazil?' The analysis_tool has provided a
detailed breakdown of the drivers of salience for Corona Extra in Brazil, including both physical and mental
availability drivers, with specific metrics and changes over time. The summary clearly identifies physical
availability as the primary driver, with supporting data on trade execution, pack availability, trade visibility,
price perception, spontaneous awareness, and brand clarity. This directly addresses the user's question with relevant
and sufficient detail. No additional information is required to answer the question as posed.",
"missing_information": ""}, "iteration": 1,
 "next_tool_timeout": 60}
```


8.Final answer tool


```
{"node": "final_answer_tool", "next_tool": "END", "question": "What are the Drivers of Salience for Corona in
Brazil?", "outcome": "# Drivers of Salience for Corona in Brazil\n\n## Overview\nThe salience of Corona Extra in
Brazil increased from **93.6** in 2024 to **100.6** in 2025 (+7.0 p.p.). The primary driver of this growth is
**physical availability**.\n\n## Key Drivers\n\n### Physical Availability (Primary Driver)\n- **Trade Execution
(P4W):** Improved from 21.3 to 24.1 (+2.8 p.p.)\n- **Pack Availability:** Increased from 20.5 to 22.0 (+1.5 p.p.)\n-
**Trade Visibility:** Rose significantly from 86.6 to 97.7 (+11.1 p.p.)\n- **Price Perception:** Slight decline from
24.4 to 23.8 (-0.6 p.p.)\n\n### Mental Availability\n- **Spontaneous Awareness:** Increased from 18.8 to 22.1 (+3.3
p.p.)\n- **Brand Clarity:** Slight improvement from 3.2 to 3.3 (+0.1 p.p.)\n\n## Data Summary\n\n| KPI
| 2024 FY FULL YEAR | 2025 FY FULL YEAR | Delta  | Notes                |\n|------------------------|-----------------
--|------------------|--------|----------------------|\n| Salience               | 93.6             | 100.6
| +7.0   | Driven by physical availability |\n| Trade Execution (P4W)  | 21.3             | 24.1            | +2.8
| Physical availability |\n| Pack Availability      | 20.5             | 22.0            | +1.5   | Physical
availability |\n| Trade Visibility       | 86.6             | 97.7            | +11.1  | Physical availability |\n|
Price Perception       | 24.4             | 23.8            | -0.6   | Physical availability |\n| Spontaneous
Awareness  | 18.8             | 22.1            | +3.3   | Mental availability   |\n| Brand Clarity          | 3.2
| 3.3              | +0.1   | Mental availability   |\n\n## Conclusion\nCorona Extra's salience in Brazil is
primarily driven by improvements in **physical availability**\u2014notably trade execution, pack availability, and
trade visibility. Mental availability factors such as spontaneous awareness and brand clarity also contributed, but
to a lesser extent.", "next_tool_timeout": 0}
```

Gracefully Handling Connection Issues

Inactivity Detection (5 min) - Recovers stale connections on tab switch **Sort-Key Deduplication** - Prevents duplicate message processing **syncNextEvent Mutation** - Recovers dropped messages based on timeouts **Custom Event System** - Decouples recovery logic from PubSub client with `syncNextEvent` mutation **Graceful Connection Replacement** - Stops old client before starting new one

Data Flow

1. negotiatePubSub() → Get connection URL with JWT
2. WebPubSubClient() → Connect with auto-reconnect
3. 'server-message' event → Process PubSubEventData
4. Sort-key check → Deduplicate messages
5. Redux state update + Notification dispatch
6. Tool timeout → syncNextEvent if needed
7. Custom 'TOOL_EVENT_RECOVERED' event → Process recovered message

Message Recovery & NextEvent Mutation

When messages are dropped (network issues, tab backgrounded, etc.), the frontend needs a recovery mechanism to fetch missed events.

IC Handles this with a `syncNextEvent` mutation, where frontend detected a potential drop (e.g., no events received within expected `next_tool_timeout`)it automatically calls `syncNextEvent`.

- On cases of **ACCEPTED** > timeout +3s OR **PENDING status**> `next_tool_timeout`+3s

1. Triggers `syncNextEvent` mutation via POST /v2/conversations/{request_id}/sync with the `request_id` and `after_id` (last received event ID)
2. /sync endpoint returns all events after the `after_id`, allowing the frontend to catch up on missed messages
3. Dispatches `TOOL_EVENT_RECOVERED` custom event based on the response, which includes any missed events
4. ChatLayout listens and processes like normal PubSub message
5. Incase of no new events , /sync updates the event as `ABANDONED` which the frontend can use to show appropriate message to user and stop waiting for events

IC UI Code Reference:

- src/hooks/useWebPubSub.tsx
- src/hooks/useConversationNotifications.tsx
- src/components/pages/Messages/MessageContent/useToolTimeout.tsx
- src/services/apiService/index.ts
- src/services/apiService/globals.ts